Methods

Table 1: Simulation mis-specification design matrix. Outer dimensions describe the unique correct models (6) while inner dimensions (18) describe the unique mis-specifications run for each simulation.

Covariance Matrix	**Autocorrelation**	**Matern Correlation**	**Phylogenetic Correlation**
Type	Mis-specification		
LMM	Data Model iid Mis-specified Covariance	Data Model iid Mis-specified Covariance	Data Model iid Mis-specified Covariance
GLMM	Data Model iid Mis-specified Covariance	Data Model iid Mis-specified Covariance	Data Model iid Mis-specified Covariance

Table 2: Linear Model Simulation: data generating models, parameter values, and mis-specifications.

	Data Generating Model	Parameters	Data Fitting Model
Correct	$X_i \sim N(0,1)$ $\mu_{i,j} = X_i\beta$ $y_{i,j} \sim N(\mu_{i,j}, \sigma_y)$	$\beta = (4, -5)$ $\sigma_y = 1$	$X_i \sim N(0,1)$ $\mu_{i,j} = X_i\beta$ $y_{i,j} \sim N(\mu_{i,j}, \sigma_y)$
Mis-specified	$X_i \sim N(0,1)$ $\mu_{i,j} = X_i\beta$ $y'_{i,j} \sim N(\mu_{i,j}, exp(\sigma_y))$		$X_i \sim N(0,1)$ $\mu_{i,j} = X_i\beta$ $y'_{i,j} \sim N(\mu_{i,j}, \sigma_y)$

Table 3: Temporal Model Simulation: data generating models, parameter values, and mis-specifications.

	Linear Mixed Model Data Generating Model	Parameters	Data Fitting Model	**Generalized Linear Mixed Model** Data Generating Model	Parameters	Data Fitting Model
Correct	$\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i \sim N(u_i, \sigma_y)$	$a = 2$ $u[1] = 0$ $\sigma_u = 1$ $\sigma_y = 1$	$\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i \sim N(u_i, \sigma_y)$	$\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i \sim Gamma(\frac{1}{CV}^2, e^{u_i}CV^2)$	$a = .02$ $u[1] = 0$ $\sigma_u = 0.1$ $CV = 0.3$	$\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i \sim Gamma(\frac{1}{CV}^2, e^{u_i}CV^2)$
Mis-specified	Missing Random Effects $\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i \sim N(u_i, \sigma_y)$		$y_i \sim N(a(1:n), \sigma_y)$	Missing Random Effects $\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i \sim Gamma(\frac{1}{CV}^2, e^{u_i}CV^2)$		$y_i \sim Gamma(\frac{1}{CV}^2, e^{a}CV^2)$
	Mis-specified Data Model $\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $\sigma_y^2 = c(rep(35, n/4), rep(0.5, n/4),$ $rep(35, n/4), rep(0.5, n/4))$ $y_i^{'} \sim N(u_i, \sigma_y)$		$\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i^{'} \sim N(u_i, \sigma_y)$	Misp-specified Data Model $\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i \sim Gamma(\frac{1}{CV}^2, e^{u_i}CV^2)$		$\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i \sim N(u_i, \sigma_y)$
	Mis-specifed Random Effects $\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i \sim N(u_i, \sigma_y)$		$\mu_i = u_{i-1}$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i \sim N(u_i, \sigma_y)$	Mis-specified Random Effects $u_i^{'} = u_{i-1} + Gamma(0.5, 10)$ $y_i^{'} \sim Gamma(\frac{1}{CV}^2, e^{u_i^{'}}CV^2)$		$\mu_i = u_{i-1} + a$ $u_i \sim N(\mu_i, \sigma_u)$ $y_i^{'} \sim Gamma(\frac{1}{CV}^2, e^{u_i}CV^2)$

Table 4: Spatial Model Simulation: data generating models, parameter values, and mis-specifications.

	Linear Mixed Model Data Generating Model	Parameters	Data Fitting Model	**Generalized Linear Mixed Model** Data Generating Model	Parameters	Data Fitting Model
Correct	$\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = \beta_0 + \omega_i$ $y \sim N(\eta, \sigma_y)$	$\beta_0 = 2$ $\sigma^2_y = 1$ $\phi = 50$ $\kappa = \sqrt{8}/\phi$ $\sigma^2_\omega = 2$	$\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = \beta_0 + \omega_i$ $y \sim N(\eta, \sigma_y)$	$\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = exp(\beta_0 + \omega_i)$ $y \sim Pois(\eta, \sigma_y)$	$\beta = 2$ $\phi = 30$ $\kappa = \sqrt{8}/\phi$ $\sigma^2_\omega = 0.25$	$\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = exp(\beta_0 + \omega_i)$ $y \sim Pois(\eta, \sigma_y)$
Mis-specified	Missing Random Effects $\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = \beta_0 + \omega_i$ $y \sim N(\eta, \sigma_y)$		$\eta_i = \beta_0$ $y \sim N(\eta, \sigma_y)$	Missing Random Effects $\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = exp(\beta_0 + \omega_i)$ $y \sim Pois(\eta)$		$\eta_i = exp(\beta_0)$ $y \sim Pois(\eta)$
	Mis-specified Data Model $\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = \beta_0 + \omega_i$ $y \sim N(\eta, \sigma_y)$		$\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = \beta_0 + \omega_i$ $\sigma_y = exp(N(0,1))$ $y \sim N(\eta, \sigma_y)$	Mis-specified Data Model $\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = exp(\beta_0 + \omega_i)$ $y^{'} \sim B(1, 0.7) * Pois(\eta)$		$\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = exp(\beta_0 + \omega_i)$ $y \sim Pois(\eta)$
	Mis-specified Random Effects Model $\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = \beta_0 + exp(\omega_i)$ $y^{'} \sim N(\eta, \sigma_y)$		$\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = \beta_0 + \omega_i$ $y^{'} \sim N(\eta, \sigma_y)$	Mis-specified Random Effects Model $R = 45°; S = \sqrt(15)$ $Loc_{aniso} = (Loc)S^{-1}R^{-1}$ $\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = \beta_0 + \omega_i$ $y^{'} \sim N(\eta, \sigma_y)$		$\omega \sim GMRF(Q[\kappa, \sigma^2_\omega])$ $\eta_i = \beta_0 + \omega_i$ $y^{'} \sim N(\eta, \sigma_y)$

Table 5: Phylogenetic Model Simulation: data generating models, parameter values, and mis-specifications.

	Linear Mixed Model			**Generalized Linear Mixed Model**		
	Data Generating Model	Parameters	Data Fitting Model	Data Generating Model	Parameters	Data Fitting Model
Correct	$tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = X\beta + u$ $y \sim Normal(\eta, \sigma_y)$	$a = 0$ $r = 0$ $\sigma_u^2 = 1$ $\beta = (0.5, -2)$ $\sigma_y = 1$	$tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = X\beta + u$ $y \sim Normal(\eta, \sigma_y)$	$tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = exp(X\beta + u)$ $y \sim Poisson(\eta)$	$a = 0$ $r = 0$ $\sigma_u^2 = 1$ $\beta = (0.5, -2)$	$tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = exp(X\beta + u)$ $y \sim Poisson(\eta)$
Mis-specified	Missing Random Effects $tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = X\beta + u$ $y \sim Normal(\eta, \sigma_y)$		$X_i \sim Unif(-0.5, 0.5)$ $\eta = X\beta$ $y \sim Normal(\eta, \sigma_y)$	Missing Random Effects $tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = exp(X\beta + u)$ $y \sim Poisson(\eta)$		$X_i \sim Unif(-0.5, 0.5)$ $\eta = exp(X\beta)$ $y \sim Poisson(\eta)$
	Mis-specified Data Model $tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = X\beta + u$ $y \sim Normal(\eta, \sigma_y)$		$tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = exp(X\beta + u)$ $y \sim Normal(\eta, \sigma_y)$	Misp-specified Data Model $tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = exp(X\beta + u)$ $y \sim Poisson(\eta)$		$tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $\eta = exp(\beta_0 + u)$ $y \sim Poisson(\eta)$
	Mis-specifed Random Effects $tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = X\beta + exp(u)$ $y^{'} \sim Normal(\eta, \sigma_y)$		$tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = X\beta + u$ $y^{'} \sim Normal(\eta, \sigma_y)$	Mis-specified Random Effects $tree \sim randomTree(n)$ $\Sigma = EB(tree, a = 1, r = -2, \sigma_u^2)$ $u^{'} \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta^{'} = exp(X\beta + u^{'})$ $y^{'} \sim Poisson(\eta^{'})$		$tree \sim randomTree(n)$ $\Sigma = BM(tree, a, r, \sigma_u^2)$ $u \sim MVNORM(\Sigma)$ $X_i \sim Unif(-0.5, 0.5)$ $\eta = exp(X\beta + u)$ $y^{'} \sim Poisson(\eta)$

Results

LM

Table 6: Linear Model. Type I error rates at the 0.05 significance level evaluated for each method for theoretical and estimated residuals.

test	residual type	Pearson	one-step cdf	one-step Generic	one-step Gaussian	full Gaussian
Kolmogorov-Smirnov	theoretical	0.048	0.048	0.048	0.048	0.048
	estimated	0.000	0.000	0.000	0.000	0.000

Table 7: Linear Model. Power at the 0.95 significance level evaluated for each method for theoretical and estimated residuals.

test	residual type	Pearson	one-step cdf	one-step Generic	one-step Gaussian	full Gaussian
Kolmogorov-Smirnov	theoretical	1.000	1.000	1.000	1.000	1.000
	estimated	0.963	0.963	0.963	0.963	0.963

Temporal Correlation - LMM

Table 8: LMM Temporal Correlation Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right), test type and method (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	B: Heterosck.	C: Missing Drift
Anderson-Darling	Pearson	0.061	1.000	1.000	1.000
Anderson-Darling	one-step cdf	0.046	1.000	1.000	1.000
Anderson-Darling	one-step Generic	0.046	1.000	1.000	1.000
Anderson-Darling	one-step Gaussian	0.046	1.000	1.000	1.000
Anderson-Darling	full Gaussian	0.046	1.000	1.000	1.000
Anderson-Darling	MCMC	0.049	1.000	1.000	0.051
Anderson-Darling	Unconditional ecdf	0.335	1.000	1.000	1.000
Anderson-Darling	Conditional ecdf	0.228	1.000	1.000	1.000
Kolmogorov- Smirnov	Pearson	0.056	1.000	1.000	1.000
Kolmogorov- Smirnov	one-step cdf	0.041	1.000	0.889	1.000
Kolmogorov- Smirnov	one-step Generic	0.042	1.000	0.867	1.000
Kolmogorov- Smirnov	one-step Gaussian	0.042	1.000	0.867	1.000
Kolmogorov- Smirnov	full Gaussian	0.042	1.000	0.867	1.000
Kolmogorov- Smirnov	MCMC	0.039	1.000	0.813	0.045
Kolmogorov- Smirnov	Unconditional ecdf	0.060	1.000	1.000	1.000
Kolmogorov- Smirnov	Conditional ecdf	0.057	1.000	1.000	1.000
Autocorrelation	Pearson	0.047	1.000	0.000	1.000
Autocorrelation	one-step cdf	0.051	1.000	0.000	0.100
Autocorrelation	one-step Generic	0.052	1.000	0.000	0.104
Autocorrelation	one-step Gaussian	0.052	1.000	0.000	0.100
Autocorrelation	full Gaussian	0.052	1.000	0.000	0.100
Autocorrelation	MCMC	0.048	1.000	0.000	0.075
Autocorrelation	Unconditional ecdf	0.043	0.982	0.000	0.132
Autocorrelation	Conditional ecdf	0.049	0.983	0.000	0.986

Table 9: LMM Temporal Correlation Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right), test type and method (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	B: Heterosck.	C: Missing Drift
Anderson-Darling	Pearson	0.002	1.000	0.010	0.000
Anderson-Darling	one-step cdf	0.030	1.000	0.016	1.000
Anderson-Darling	one-step Generic	0.024	1.000	0.016	0.995
Anderson-Darling	one-step Gaussian	0.022	1.000	0.014	0.996
Anderson-Darling	full Gaussian	0.027	1.000	0.022	0.997
Anderson-Darling	MCMC	0.057	1.000	0.121	0.051
Anderson-Darling	Unconditional ecdf	0.315	1.000	0.912	0.999
Anderson-Darling	Conditional ecdf	0.010	1.000	0.672	0.000
Kolmogorov- Smirnov	Pearson	0.129	1.000	0.667	1.000
Kolmogorov- Smirnov	one-step cdf	0.000	1.000	0.362	1.000
Kolmogorov- Smirnov	one-step Generic	0.000	1.000	0.364	1.000
Kolmogorov- Smirnov	one-step Gaussian	0.000	1.000	0.364	1.000
Kolmogorov- Smirnov	full Gaussian	0.000	1.000	0.370	1.000
Kolmogorov- Smirnov	MCMC	0.059	1.000	0.471	0.048
Kolmogorov- Smirnov	Unconditional ecdf	0.001	1.000	0.142	1.000
Kolmogorov- Smirnov	Conditional ecdf	0.129	1.000	0.657	1.000
Lilliefors	Pearson	0.051	0.130	0.992	0.022
Lilliefors	one-step cdf	0.048	0.130	0.980	1.000
Lilliefors	one-step Generic	0.052	0.130	0.980	0.036
Lilliefors	one-step Gaussian	0.053	0.130	0.980	0.036
Lilliefors	full Gaussian	0.052	0.130	0.982	0.034
Lilliefors	MCMC	0.048	0.130	0.872	0.053
Lilliefors	Unconditional ecdf	0.169	0.204	0.970	0.141
Lilliefors	Conditional ecdf	0.051	0.289	0.995	0.078
Autocorrelation	Pearson	0.001	1.000	0.000	0.000
Autocorrelation	one-step cdf	0.003	1.000	0.000	0.001
Autocorrelation	one-step Generic	0.003	1.000	0.000	0.000
Autocorrelation	one-step Gaussian	0.003	1.000	0.000	0.000
Autocorrelation	full Gaussian	0.003	1.000	0.000	0.000
Autocorrelation	MCMC	0.025	1.000	0.010	0.054
Autocorrelation	Unconditional ecdf	0.006	1.000	0.177	0.000
Autocorrelation	Conditional ecdf	0.001	1.000	0.048	0.051

Temporal Correlation - GLMM

Table 10: GLMM Temporal Correlation Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	B: Gamma - Normal	C: Misp Cov
Anderson-Darling	Pearson	0.976	1.000	1.000	1.000
Anderson-Darling	one-step cdf	0.097	1.000	0.961	1.000
Anderson-Darling	one-step Generic	0.041	1.000	0.959	1.000
Anderson-Darling	MCMC	0.044	1.000	0.688	0.051
Anderson-Darling	Conditional ecdf	0.227	1.000	1.000	1.000
Kolmogorov- Smirnov	Pearson	0.875	1.000	1.000	1.000
Kolmogorov- Smirnov	one-step cdf	0.077	1.000	0.874	1.000
Kolmogorov- Smirnov	one-step Generic	0.033	1.000	0.880	1.000
Kolmogorov- Smirnov	MCMC	0.035	1.000	0.580	0.050
Kolmogorov- Smirnov	Conditional ecdf	0.049	1.000	1.000	1.000
Autocorrelation	Pearson	0.217	1.000	1.000	1.000
Autocorrelation	one-step cdf	0.054	0.995	0.331	0.000
Autocorrelation	one-step Generic	0.053	0.993	0.092	0.000
Autocorrelation	MCMC	0.044	1.000	0.024	0.134
Autocorrelation	Conditional ecdf	0.051	0.975	NA	0.986

Table 11: GLMM Temporal Correlation Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right), test type and method (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	B: Gamma - Normal	C: Misp Cov
Anderson-Darling	Pearson	0.622	0.912	0.001	1.000
Anderson-Darling	one-step cdf	0.115	0.831	0.292	0.023
Anderson-Darling	one-step Generic	0.038	0.883	0.018	0.021
Anderson-Darling	MCMC	0.049	0.839	0.052	0.039
Anderson-Darling	Conditional ecdf	0.000	0.862	0.223	0.042
Kolmogorov- Smirnov	Pearson	0.878	0.986	0.982	1.000
Kolmogorov- Smirnov	one-step cdf	0.115	0.958	0.797	0.002
Kolmogorov- Smirnov	one-step Generic	0.000	0.965	0.795	0.002
Kolmogorov- Smirnov	MCMC	0.044	0.958	0.091	0.049
Kolmogorov- Smirnov	Conditional ecdf	0.924	0.950	0.979	0.039
Lilliefors	Pearson	0.993	0.983	0.966	1.000
Lilliefors	one-step cdf	0.184	0.779	0.956	0.126
Lilliefors	one-step Generic	0.049	0.779	0.961	0.123
Lilliefors	MCMC	0.051	0.779	0.163	0.051
Lilliefors	Conditional ecdf	0.051	0.871	0.696	0.077
Autocorrelation	Pearson	0.004	1.000	0.021	0.034
Autocorrelation	one-step cdf	0.004	1.000	0.107	0.004
Autocorrelation	one-step Generic	0.002	1.000	0.122	0.005
Autocorrelation	MCMC	0.038	1.000	0.081	0.024
Autocorrelation	Conditional ecdf	0.000	0.999	0.036	0.000

Spatial

Table 12: LMM Spatial Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right), test type and method (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.044	0.999	1.000	1.000
Anderson-Darling	one-step cdf	0.009	0.999	1.000	0.881
Anderson-Darling	one-step Generic	0.011	0.999	1.000	0.778
Anderson-Darling	one-step Gaussian	0.011	0.999	1.000	0.889
Anderson-Darling	full Gaussian	0.011	0.999	1.000	0.889
Anderson-Darling	MCMC	0.057	0.999	1.000	0.757
Anderson-Darling	Unconditional ecdf	0.225	1.000	1.000	0.974
Anderson-Darling	Conditional ecdf	0.216	1.000	1.000	1.000
Kolmogorov- Smirnov	Pearson	0.046	0.992	0.977	1.000
Kolmogorov- Smirnov	one-step cdf	0.012	0.992	0.978	0.536
Kolmogorov- Smirnov	one-step Generic	0.012	0.992	0.973	0.538
Kolmogorov- Smirnov	one-step Gaussian	0.012	0.992	0.973	0.538
Kolmogorov- Smirnov	full Gaussian	0.012	0.992	0.973	0.538
Kolmogorov- Smirnov	MCMC	0.047	0.992	0.961	0.480
Kolmogorov- Smirnov	Unconditional ecdf	0.016	0.996	0.992	0.649
Kolmogorov- Smirnov	Conditional ecdf	0.039	0.992	0.980	1.000
Autocorrelation	Pearson	0.059	0.998	0.148	0.958
Autocorrelation	one-step cdf	0.050	0.998	0.014	0.182
Autocorrelation	one-step Generic	0.050	0.998	0.012	0.334
Autocorrelation	one-step Gaussian	0.050	0.998	0.011	0.314
Autocorrelation	full Gaussian	0.050	0.998	0.011	0.314
Autocorrelation	MCMC	0.076	0.998	0.001	0.142
Autocorrelation	Unconditional ecdf	0.045	0.996	0.013	0.130
Autocorrelation	Conditional ecdf	0.054	0.998	0.057	0.515

Table 13: LMM Spatial Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right), test type and method (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.002	0.034	0.053	0.199
Anderson-Darling	one-step cdf	0.036	0.022	0.052	0.193
Anderson-Darling	one-step Generic	0.043	0.037	NA	NA
Anderson-Darling	one-step Gaussian	0.041	0.035	0.053	0.203
Anderson-Darling	full Gaussian	0.041	0.040	0.055	0.213
Anderson-Darling	MCMC	0.037	0.034	0.058	0.200
Anderson-Darling	Unconditional ecdf	0.305	0.257	0.999	0.974
Anderson-Darling	Conditional ecdf	0.017	0.151	0.994	0.872
Kolmogorov- Smirnov	Pearson	0.137	0.004	0.985	0.936
Kolmogorov- Smirnov	one-step cdf	0.017	0.004	0.982	0.871
Kolmogorov- Smirnov	one-step Generic	0.013	0.004	NA	NA
Kolmogorov- Smirnov	one-step Gaussian	0.013	0.004	0.983	0.869
Kolmogorov- Smirnov	full Gaussian	0.013	0.004	0.988	0.882
Kolmogorov- Smirnov	MCMC	0.039	0.004	0.984	0.843
Kolmogorov- Smirnov	Unconditional ecdf	0.026	0.002	0.931	0.851
Kolmogorov- Smirnov	Conditional ecdf	0.144	0.002	0.986	0.937
Lilliefors	Pearson	0.053	0.124	0.999	0.918
Lilliefors	one-step cdf	0.052	0.124	0.999	0.943
Lilliefors	one-step Generic	0.049	0.122	NA	NA
Lilliefors	one-step Gaussian	0.049	0.124	0.999	0.947
Lilliefors	full Gaussian	0.049	0.124	1.000	0.963
Lilliefors	MCMC	0.050	0.124	0.999	0.863
Lilliefors	Unconditional ecdf	0.144	0.196	1.000	0.970
Lilliefors	Conditional ecdf	0.064	0.171	0.999	0.920
Autocorrelation	Pearson	0.004	0.998	0.753	0.852
Autocorrelation	one-step cdf	0.045	0.998	0.737	0.858
Autocorrelation	one-step Generic	0.046	0.998	NA	NA
Autocorrelation	one-step Gaussian	0.046	0.998	0.737	0.860
Autocorrelation	full Gaussian	0.046	0.998	0.762	0.880
Autocorrelation	MCMC	0.036	0.998	0.753	0.857
Autocorrelation	Unconditional ecdf	0.039	0.998	0.821	0.850
Autocorrelation	Conditional ecdf	0.004	0.998	0.859	0.848

Table 14: GLMM Spatial Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right), test type and method (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.099	0.995	1.000	0.099
Anderson-Darling	one-step cdf	0.040	0.991	1.000	0.140
Anderson-Darling	one-step Generic	0.037	0.991	1.000	0.130
Anderson-Darling	MCMC	0.062	0.991	1.000	0.070
Anderson-Darling	Conditional ecdf	0.139	0.997	1.000	0.128
Kolmogorov- Smirnov	Pearson	0.085	0.979	1.000	0.085
Kolmogorov- Smirnov	one-step cdf	0.040	0.916	1.000	0.084
Kolmogorov- Smirnov	one-step Generic	0.037	0.916	1.000	0.083
Kolmogorov- Smirnov	MCMC	0.055	0.912	0.993	0.049
Kolmogorov- Smirnov	Conditional ecdf	0.052	0.912	1.000	0.057
Autocorrelation	Pearson	0.064	0.988	0.094	0.064
Autocorrelation	one-step cdf	0.055	0.990	0.000	0.620
Autocorrelation	one-step Generic	0.055	0.989	0.001	0.618
Autocorrelation	MCMC	0.066	0.988	0.000	0.273
Autocorrelation	Conditional ecdf	0.063	0.985	0.060	0.062

Table 15: GLMM Spatial Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right), test type and method (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.005	0.829	0.007	0.003
Anderson-Darling	one-step cdf	0.027	0.751	0.221	0.049
Anderson-Darling	one-step Generic	0.036	0.759	0.129	0.038
Anderson-Darling	MCMC	0.048	0.753	0.100	0.038
Anderson-Darling	Conditional ecdf	0.009	0.970	0.005	0.006
Kolmogorov- Smirnov	Pearson	0.253	0.984	1.000	0.555
Kolmogorov- Smirnov	one-step cdf	0.006	0.820	0.931	0.001
Kolmogorov- Smirnov	one-step Generic	0.007	0.820	0.745	0.001
Kolmogorov- Smirnov	MCMC	0.027	0.820	0.082	0.024
Kolmogorov- Smirnov	Conditional ecdf	0.144	0.813	1.000	0.473
Lilliefors	Pearson	0.062	0.972	1.000	0.129
Lilliefors	one-step cdf	0.047	0.309	0.996	0.044
Lilliefors	one-step Generic	0.047	0.270	0.994	0.039
Lilliefors	MCMC	0.054	0.318	0.060	0.052
Lilliefors	Conditional ecdf	0.153	0.697	1.000	0.293
Autocorrelation	Pearson	0.001	0.988	0.001	0.348
Autocorrelation	one-step cdf	0.079	0.990	0.009	0.829
Autocorrelation	one-step Generic	0.079	0.989	0.008	0.824
Autocorrelation	MCMC	0.059	0.989	0.036	0.316
Autocorrelation	Conditional ecdf	0.001	0.989	0.004	0.335

Phylogenetic

Table 16: LMM Phylogenetic Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right, test type and method (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.055	1.000	1.000	1.000
Anderson-Darling	one-step cdf	0.048	1.000	0.975	0.905
Anderson-Darling	one-step Generic	0.048	1.000	0.926	0.807
Anderson-Darling	one-step Gaussian	0.048	1.000	0.981	0.914
Anderson-Darling	full Gaussian	0.048	1.000	0.981	0.914
Anderson-Darling	MCMC	0.046	1.000	0.945	0.809
Anderson-Darling	Unconditional ecdf	0.318	1.000	0.997	0.983
Anderson-Darling	Conditional ecdf	0.229	1.000	1.000	1.000
Kolmogorov- Smirnov	Pearson	0.047	0.998	1.000	1.000
Kolmogorov- Smirnov	one-step cdf	0.043	0.998	0.785	0.546
Kolmogorov- Smirnov	one-step Generic	0.042	0.998	0.779	0.544
Kolmogorov- Smirnov	one-step Gaussian	0.042	0.998	0.779	0.544
Kolmogorov- Smirnov	full Gaussian	0.042	0.998	0.779	0.544
Kolmogorov- Smirnov	MCMC	0.044	0.998	0.806	0.586
Kolmogorov- Smirnov	Unconditional ecdf	0.048	0.998	0.868	0.689
Kolmogorov- Smirnov	Conditional ecdf	0.043	0.998	1.000	1.000
Autocorrelation	Pearson	0.045	0.999	0.866	0.909
Autocorrelation	one-step cdf	0.044	0.999	0.491	0.392
Autocorrelation	one-step Generic	0.045	0.999	0.234	0.272
Autocorrelation	one-step Gaussian	0.045	0.999	0.343	0.328
Autocorrelation	full Gaussian	0.045	0.999	0.343	0.328
Autocorrelation	MCMC	0.045	0.999	0.375	0.254
Autocorrelation	Unconditional ecdf	0.039	0.988	0.073	0.114
Autocorrelation	Conditional ecdf	0.051	0.990	0.317	0.376

Table 17: LMM Phylogenetic Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right), test type and method (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.000	0.034	0.098	0.050
Anderson-Darling	one-step cdf	0.046	0.040	0.453	0.486
Anderson-Darling	one-step Generic	0.037	0.038	0.005	0.009
Anderson-Darling	one-step Gaussian	0.026	0.035	0.284	0.229
Anderson-Darling	full Gaussian	0.031	0.028	0.281	0.237
Anderson-Darling	MCMC	0.047	0.037	0.106	0.062
Anderson-Darling	Unconditional ecdf	0.289	0.312	0.994	0.985
Anderson-Darling	Conditional ecdf	0.006	0.164	0.585	0.411
Kolmogorov- Smirnov	Pearson	0.271	0.002	0.987	0.944
Kolmogorov- Smirnov	one-step cdf	0.013	0.002	0.950	0.864
Kolmogorov- Smirnov	one-step Generic	0.009	0.002	0.966	0.884
Kolmogorov- Smirnov	one-step Gaussian	0.009	0.002	0.966	0.884
Kolmogorov- Smirnov	full Gaussian	0.009	0.002	0.965	0.883
Kolmogorov- Smirnov	MCMC	0.038	0.002	0.368	0.217
Kolmogorov- Smirnov	Unconditional ecdf	0.017	0.002	0.910	0.813
Kolmogorov- Smirnov	Conditional ecdf	0.282	0.002	0.985	0.943
Lilliefors	Pearson	0.052	0.105	0.994	0.941
Lilliefors	one-step cdf	0.046	0.105	0.986	0.956
Lilliefors	one-step Generic	0.044	0.105	0.987	0.944
Lilliefors	one-step Gaussian	0.044	0.105	0.993	0.965
Lilliefors	full Gaussian	0.044	0.105	0.993	0.965
Lilliefors	MCMC	0.050	0.105	0.525	0.355
Lilliefors	Unconditional ecdf	0.163	0.191	0.996	0.984
Lilliefors	Conditional ecdf	0.059	0.184	0.699	0.611
Autocorrelation	Pearson	0.824	0.998	0.786	0.821
Autocorrelation	one-step cdf	0.015	0.998	0.306	0.210
Autocorrelation	one-step Generic	0.015	0.998	0.364	0.275
Autocorrelation	one-step Gaussian	0.015	0.998	0.383	0.314
Autocorrelation	full Gaussian	0.015	0.998	0.376	0.312
Autocorrelation	MCMC	0.035	0.998	0.180	0.081
Autocorrelation	Unconditional ecdf	0.019	0.993	0.262	0.164
Autocorrelation	Conditional ecdf	0.802	0.998	0.430	0.411

Table 18: GLMM Phylogenetic Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right), test type and method (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	C: Misp Cov	NA
Anderson-Darling	Pearson	0.531	1.000	1.000	1.000
Anderson-Darling	one-step cdf	0.112	1.000	0.971	0.234
Anderson-Darling	one-step Generic	0.023	1.000	0.969	0.094
Anderson-Darling	MCMC	0.049	1.000	0.271	0.059
Anderson-Darling	Conditional ecdf	0.091	1.000	1.000	0.964
Kolmogorov- Smirnov	Pearson	0.536	1.000	1.000	0.928
Kolmogorov- Smirnov	one-step cdf	0.098	0.998	0.948	0.140
Kolmogorov- Smirnov	one-step Generic	0.026	0.997	0.926	0.052
Kolmogorov- Smirnov	MCMC	0.044	0.997	0.230	0.041
Kolmogorov- Smirnov	Conditional ecdf	0.031	0.999	1.000	0.597
Autocorrelation	Pearson	0.041	0.976	0.952	0.157
Autocorrelation	one-step cdf	0.056	0.984	0.308	0.090
Autocorrelation	one-step Generic	0.039	0.989	0.316	0.085
Autocorrelation	MCMC	0.042	0.991	0.084	0.064
Autocorrelation	Conditional ecdf	0.051	0.983	0.905	0.059

Table 19: GLMM Phylogenetic Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right), test type and method (top to bottom).

		Type I Error	Power		
test	method	Correct	A: iid	C: Misp Cov	NA
Anderson-Darling	Pearson	0.004	0.998	0.012	0.003
Anderson-Darling	one-step cdf	0.079	0.985	0.027	0.101
Anderson-Darling	one-step Generic	0.041	0.983	0.022	0.040
Anderson-Darling	MCMC	0.056	0.985	0.047	0.052
Anderson-Darling	Conditional ecdf	0.001	1.000	0.031	0.001
Kolmogorov- Smirnov	Pearson	1.000	1.000	0.150	1.000
Kolmogorov- Smirnov	one-step cdf	0.061	0.996	0.017	0.091
Kolmogorov- Smirnov	one-step Generic	0.012	0.996	0.018	0.014
Kolmogorov- Smirnov	MCMC	0.028	0.996	0.033	0.055
Kolmogorov- Smirnov	Conditional ecdf	0.855	0.995	0.077	0.919
Lilliefors	Pearson	0.662	1.000	0.164	0.826
Lilliefors	one-step cdf	0.059	0.962	0.076	0.068
Lilliefors	one-step Generic	0.024	0.967	0.081	0.030
Lilliefors	MCMC	0.050	0.988	0.063	0.054
Lilliefors	Conditional ecdf	0.796	0.999	0.131	0.865
Autocorrelation	Pearson	0.621	0.975	0.240	0.863
Autocorrelation	one-step cdf	0.063	0.967	0.158	0.091
Autocorrelation	one-step Generic	0.048	0.985	0.160	0.097
Autocorrelation	MCMC	0.051	0.988	0.068	0.055
Autocorrelation	Conditional ecdf	0.360	0.977	0.224	0.530

Table 20: Overview of issues and recommendations for common classes of models. Correlation and distributions refer to predicted data from a fitted model, against which observed points are compared. A linear rotation refers to a multiplication of the simulated and observed data by a Cholesky decomposition of the estimated covariance matrix of the observed data, z'=Lz, as available in DHARMa.

Model class	Case studies	Issues and causes	Recommendation
Linear model	Linear model	No issues	Pearson residuals
Generalized linear model (GLM)	Skewed Gamma	Non-normality resulting from response variable. Quantile residuals are needed if not approximately normal.	Quantile residual
Linear mixed model (LMM), Multivariate model	Random walk, Spatial LMM, Multinomial	Linear correlations caused by non-independence in observations.	Use a method that linearly decorrela order to transform to a unit iid no OSA Full Gaussian, OSA one-step sian, or simulation residuals with rot
Generalized linear mixed model (GLMM)	Spatial Poisson, Repeated measures Tweedie	Non-normality and non-linear correlations caused by response variable and non-independence in observations.	Needs non-linear decorrelation and tiles. Needs non-linear decorrelation quantiles. Best approach depends o study and sample size.